

Executive Summary

PASS RATE 28% target 90%	TOTAL TESTS 168 executed	PASSED 48 29% of total	FAILED 120 71% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 28% | Target: 90% | Gap: **+62%**

The Revenue Management System Kenya backend API testing reveals significant stability concerns with a 28% overall pass rate against the 90% target, indicating widespread issues across both API automation (27.5% pass rate) and smoke validation (29.5% pass rate) suites. The consistent failure pattern across 120 endpoints suggests systemic backend issues rather than isolated defects, potentially involving service configuration, data handling, or infrastructure components. The low success rate indicates the backend system is not meeting expected reliability standards for production readiness, with approximately 72% of API endpoints exhibiting incorrect behavior, timeout issues, or response validation failures that require immediate technical investigation and remediation.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	87	26	61	30%	FAIL
TOTAL		168	48	120	28%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#47	04 Jul	168	28%	120	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	48	28.6%	180ms	Within SLA
Requests Failed	120	71.4%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on the 115 failing endpoints to identify common failure patterns and systemic issues	Immediate	P1
2	Prioritize fixing high-impact revenue calculation and transaction processing endpoints that affect core business operations	3 days	P1
3	Review backend service logs and monitoring data to identify performance bottlenecks, timeout issues, or resource constraints	5 days	P2
4	Implement enhanced error handling and response validation in the backend services to ensure consistent API behavior	1 week	P2
5	Establish incremental stabilization milestones targeting 50%, 70%, then 90% pass rates with focused remediation sprints	1 week	P2
6	Deploy additional backend monitoring and alerting for real-time detection of service degradation or endpoint failures	2 weeks	P3

KEY FINDINGS

1	Backend API endpoints demonstrate 72% failure rate across 168 test scenarios, indicating widespread service instability
2	Both API automation and smoke validation suites show similar failure patterns, suggesting systemic backend issues rather than isolated endpoint problems
3	Core revenue management functionalities appear compromised with only 48 endpoints returning expected responses
4	Critical gap of 62 percentage points from the 90% target pass rate indicates backend requires substantial stabilization
5	Consistent failure distribution between automation (72.5% fail) and smoke tests (70.5% fail) points to fundamental service layer issues

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation		29%
BE Smoke Validation		30%